

Question Paper

Exam Date & Time: 18-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
FIRST SEMESTER M.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

Advanced Data Structures and Algorithms [CSS_5101]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Given a k -bit binary counter, apply the potential method with a suitable potential function $\phi(i)$ to derive amortized cost for each INCREMENT. Show calculations for both actual and amortized cost for one increment. (5)
- A)

Course Outcome: "CO1" / Cognitive Level: "Application"

- B) Explain how the use of load factor-based resizing in dynamic tables helps control the overall cost of insert and delete operations, and how amortized analysis proves that the average cost remains efficient over many operations. (5)

Course Outcome: "CO1" / Cognitive Level: "Understand"

- 2) A B-Tree has order $m = 5$ and height $h = 3$. Apply the concept of B-Tree to compute: (5)

- A)
- The minimum number of keys the tree can store
 - The maximum number of keys the tree can store
 - Justify why the root node is treated differently in minimum key calculation.
 - Why the number of keys in a B-Tree is directly linked to its height?

Course Outcome: "CO2" / Cognitive Level: "Application"

- B) Given two binomial heaps H_1 and H_2 with overlapping tree degrees, analyze the sequence of structural changes that occur during UNION and explain why duplicate-degree trees cannot remain in the final heap. (5)

Course Outcome: "CO2" / Cognitive Level: "Analysis"

- 3) Apply the disjoint-set model to determine how many connected components remain after processing the graph edges: (1,2), (2,3), (4,5). (5)

- A) Use only MAKE-SET, UNION, and FIND-SET and show the set contents after all operations.

Course Outcome: "CO1" / Cognitive Level: "Application"

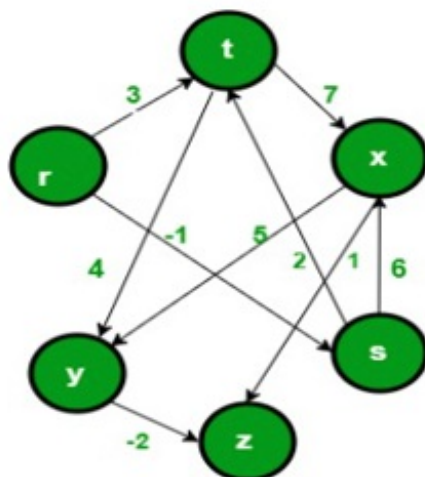
- B) Explain the meaning of the functions $\text{high}(x)$, $\text{low}(x)$, and $\text{index}(x, y)$ used in van Emde Boas trees. How do they help in navigating the recursive clusters? (5)

Course Outcome: "CO1" / Cognitive Level: "Understand"

- 4) Apply topological sorting to find single source shortest path for the following graph representing (5)

network structure. Consider the source node is 's' and show the calculated shortest path lengths.

A)



Course Outcome: "CO4" / Cognitive Level: "Application"

- B) Apply the concept of distance matrix multiplication to find the shortest paths in a complete, directed graph $G=(V, E)$ where $V = \{1, 2, 3\}$. The edge weights $w(i, j)$ are given by the initial adjacency matrix $L^{(1)}$: (5)

$$L^{(1)} = \begin{pmatrix} 0 & 3 & \infty \\ 2 & 0 & 4 \\ \infty & 1 & 0 \end{pmatrix}$$

Course Outcome: "CO4" / Cognitive Level: "Application"

- 5) Apply the concept of *spawn* and *sync* to parallelize merge sort. write the corresponding multithreaded pseudo-code and discuss its advantages. (5)

A)

Course Outcome: "CO5" / Cognitive Level: "Application"

- B) Analyze the parallelism and span of Multithreaded Matrix Multiplication and compare it with its serial counterpart. Discuss how work and critical-path length influence performance on a multicore system. (5)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

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Event Name	Subject
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058-END SEMSTER-ODD-2025-PG CSS_5101:Advanced Data Structures and Algorithms

#	Course Outcome	Description	Marks
1	1	CO 1	20
2	2	CO 2	10
3	4	CO 4	10
4	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-END SEMSTER-ODD-2025-PG CSS_5101:Advanced Data Structures and Algorithms

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	2	10(20%)
3	Application	6	30(60%)
4	Analysis	2	10(20%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
	Total	10	50 (100%)